

CUSTOMER PURCHASE BEHAVIOUR

Exploratory Data Analysis Report

~\$60	3,900	~3.7/5	~25
Avg. Spend	Total Customers	Avg. Rating	Avg. Past Orders

1. Executive Summary

This report presents a comprehensive exploratory data analysis of a customer purchase behaviour dataset containing 3,900 transaction records across four product categories, six payment methods, and 50 U.S. states.

The analysis spans demographic profiling, revenue breakdown, promotional effectiveness, loyalty patterns, and geographic distribution.

Key Finding

Across every dimension tested — age, gender, loyalty tier, discounts, promo codes, and subscriptions — no variable meaningfully predicts how much a customer spends per transaction. Average spend is anchored at ~\$60 regardless of any customer characteristic or promotional incentive applied.

The uniformity of distributions across virtually all variables is a structural fingerprint of a synthetically generated dataset.

While individual directional signals remain useful for hypothesis generation, all findings should be re-validated against real transactional data before informing strategic decisions.

2. Dataset Overview

2.1 Dataset Dimensions

The dataset consists of 3,900 customer records with 18 attributes covering demographics, purchase behaviour, product preferences, and promotional activity.

Attribute	Type	Key Values
Customer ID	Integer (identifier)	Unique per row
Age	Integer	18–70 years
Gender	Categorical	Male / Female
Location	Categorical	50 U.S. States
Purchase Amount (USD)	Numeric	\$20–\$100
Category	Categorical	Clothing, Accessories, Footwear, Outerwear
Season	Categorical	Spring, Summer, Fall, Winter
Size	Categorical	S, M, L, XL
Subscription Status	Categorical	Yes / No
Discount Applied	Categorical	Yes / No
Promo Code Used	Categorical	Yes / No
Payment Method	Categorical	6 methods
Review Rating	Numeric (1–5)	Average ~3.7
Previous Purchases	Integer	0–50
Frequency of Purchases	Categorical	Weekly to Annually

2.2 Data Quality

The dataset is complete with no missing values across all 18 columns, requiring no imputation prior to analysis.

Quality Check	Result	Notes
Missing Values	None (0)	All 3,900 rows × 18 columns complete
Duplicate Rows	None detected	Customer IDs are unique
Data Type Consistency	Passed	All columns conform to expected types
Outlier Detection	Hard bounds \$20–\$100	Likely data generation constraint, not organic

3. Demographic Analysis

3.1 Age Distribution

The age distribution is remarkably flat, with each 10-year cohort from 18–70 containing approximately 350–420 customers. Two mild peaks appear around age 20 and 45–50, but the overall uniformity is statistically improbable in real retail data.

Age Band	Approx. Count	% of Total	Observation
18–29	~780	20%	Young adults — slight concentration
30–39	~770	20%	Core working-age segment
40–49	~810	21%	Slight mid-life peak
50–59	~780	20%	Stable senior cohort
60–70	~760	19%	Under-represented but equal spenders

Analytical Note

The synthetic uniformity of age bands suppresses natural demographic correlations. Any age-stratified finding in this report should be treated as directional only.

3.2 Gender Distribution

Males constitute approximately two-thirds of the customer base, creating a pronounced gender imbalance that may affect category-level demand signals.

Gender	Count	Share
Male	2,652	68%
Female	1,248	32%

Despite this imbalance, male and female customers exhibit nearly identical median spending (~\$60) and distribution shapes, confirming that gender is not a predictor of transaction value.

3.3 Geographic Distribution

Revenue is uniformly distributed across all 50 states, ranging from ~\$5,175 (Nebraska) to ~\$5,800 (Montana) — a spread of less than 11%.

Rank	State	Total Revenue (USD)	Observation
1	Montana	~\$5,800	Highest — counterintuitive for low-population state
2	California	~\$5,700	Expected high; 2nd confirms some market effect
3–8	Various	\$5,300–\$5,600	Narrow band; no meaningful differentiation
9	New York	~\$5,275	Anomalously low for largest U.S. consumer market
10	Nebraska	~\$5,175	Lowest in top-10; still within tight range

4. Revenue & Category Analysis

4.1 Revenue by Category

Clothing is the dominant revenue engine, generating roughly 40% of total sales value. Outerwear consistently underperforms across all metrics.

Category	Revenue Tier	Key Characteristics
Clothing	Highest (~\$25K/season avg)	Clear revenue leader; summer dip notable
Accessories	Strong (~\$18K/season avg)	Highly stable; best hedging category
Footwear	Moderate (~\$9K/season avg)	No seasonal breakout; stagnant middle tier
Outerwear	Lowest (~\$4.5K/season avg)	Chronically underperforms including winter

4.2 Seasonal Revenue Heatmap Findings

Spring is the strongest overall trading season when summing across all categories. Key seasonal patterns by category:

- Clothing: Peaks in spring (\$27,692), dips in summer (\$23,078), recovers in winter (\$27,274).
- Accessories: Minimal seasonal sensitivity — revenue spans \$17,007 to \$19,874 year-round.
- Footwear: No seasonal breakout, revenue \$8,480–\$9,555.

- Outerwear: Winter revenue (\$4,562) barely exceeds summer (\$4,278) — no expected cold-season uplift.

Strategic Implication

Spring represents the optimal window for concentrated promotional and inventory investment. Outerwear's failure to peak in winter suggests a fundamental product-range or visibility problem that standard seasonal promotions alone will not fix.

4.3 Size Distribution by Category

Medium (M) is the dominant size across every category without exception. Significant supply-side concerns exist for XL and Outerwear broadly.

Category	Most Popular Size	Supply Risk	Notable Observation
Clothing	M (~775 units)	S under-stocked vs. L (285 vs. 480)	Potential lost sales for size S customers
Accessories	M (~555 units)	Balanced distribution	Healthiest size ratio across all categories
Footwear	M	Moderate	Mid-tier demand; no size extremes
Outerwear	M (<150 units)	Critical — XL barely ~30 units	All sizes dangerously low; supply/ranging issue

5. Customer Behaviour & Loyalty

5.1 Purchase Amount Distribution

Spending follows a near-uniform distribution between \$20 and \$100, with a mean of approximately \$60. The hard cut-offs at \$20 and \$100 are consistent across every subgroup examined, suggesting a predefined price tier structure in the catalogue or a data generation constraint.

Metric	Value	Interpretation
Mean Purchase Amount	~\$60	Consistent anchor across all segments
Median Purchase Amount	~\$60	Symmetric distribution — no skew
Min / Max	\$20 / \$100	Hard bounds — likely data artefact
Standard Deviation	~\$23	Moderate spread; stable revenue

Metric	Value	Interpretation
IQR (~40–80)	~\$40	Identical across all four categories

5.2 Repeat Purchase & Loyalty

Previous purchase count (ranging 0–50) has effectively zero correlation with purchase amount ($r \approx 0.04$). Loyal customers spend no more per transaction than first-time buyers.

- A substantial cohort of customers has 40–50 previous purchases, indicating strong retention.
- New customers (0–5 purchases) spend identically to veterans — no onboarding spend spike.
- Upselling cannot rely on purchase history as a trigger variable.

Business Implication

Loyalty programme value must be justified through purchase frequency and customer lifetime value metrics — not spend per visit. Basket size uplift strategies require different levers (category affinity, recency, basket composition).

6. Promotional Effectiveness

6.1 Discount Impact

Customers receiving discounts spent ~\$59 per transaction versus ~\$60 for non-discounted customers — a negligible difference with tight confidence intervals confirming the result is statistically stable.

Group	Avg. Spend (USD)	Verdict
Discount Applied = Yes	~\$59	Marginally lower spend
Discount Applied = No	~\$60	Baseline spend

The discount programme appears to erode margin without generating incremental basket value. Evaluation should shift to measuring impact on purchase frequency and customer acquisition rather than transaction size.

6.2 Promo Code Impact

Promo code users show virtually identical spend distributions to non-users across all four product categories. The only marginal signal: Footwear promo users edge ahead (~\$62 vs. ~\$59).

Category	Promo = Yes	Promo = No	Net Impact
Clothing	~\$60	~\$61	No lift — non-promo slightly higher
Accessories	~\$60	~\$61	No lift — non-promo slightly higher
Footwear	~\$62	~\$59	Marginal positive — only category signal
Outerwear	~\$57	~\$58	No lift — weakest category overall

6.3 Subscription Status

Subscribers and non-subscribers share identical median spend (~\$60) and near-identical distribution shapes. This is the third consecutive promotional variable failing to shift basket size.

Status	Median Spend	Q1	Q3
Subscribed (1,053 customers)	~\$60	~\$40	~\$80
Not Subscribed (2,847 customers)	~\$60	~\$40	~\$82

Pattern Recognition

Discounts, promo codes, and subscriptions all produce near-zero spend uplift. This consistent pattern across three independent variables strongly suggests customers operate with a fixed ~\$60 mental price anchor that no current intervention is breaking through.

7. Payment Method Analysis

All six payment methods are used with near-perfect uniformity — each capturing 16.2%–17.8% of transactions. This distribution is commercially unusual and further supports the synthetic dataset hypothesis.

Payment Method	Transaction Share	Observation
Credit Card	~17.8%	Marginal lead — not commercially dominant
PayPal	~17.1%	Digital wallet on par with physical cash

Payment Method	Transaction Share	Observation
Venmo	~16.7%	Peer-to-peer platform at retail parity — unusual
Cash	~16.6%	Equal to digital options — real-world anomaly
Debit Card	~16.2%	Joint-lowest with Bank Transfer
Bank Transfer	~16.2%	Expected for large transactions; not differentiated here

No single payment method warrants prioritised checkout optimisation investment. Full payment coverage should be maintained to avoid alienating any segment.

8. Correlation Analysis

The inter-variable correlation matrix reveals near-zero correlations across all numeric variable pairs — the single strongest statistical indicator that this dataset was synthetically generated.

Variable Pair	Correlation (r)	Interpretation
Purchase Amount vs. Review Rating	0.031	Near zero — satisfaction doesn't predict spend
Purchase Amount vs. Age	-0.010	Effectively zero — age irrelevant to spend
Purchase Amount vs. Previous Purchases	0.023	Zero — loyalty doesn't drive basket size
Age vs. Previous Purchases	0.040	Near zero — loyalty not age-stratified
Review Rating vs. Age	-0.022	Near zero — satisfaction independent of age

Statistical Caveat

A correlation range of -0.04 to +0.04 across five real-world retail variables is statistically near-impossible without deliberate randomisation. Real datasets always contain some natural covariance from confounding factors like income, geography, and product mix. All correlational findings must be re-validated on real transactional data.

9. Key Insights & Findings Summary

9.1 Confirmed Findings

- Clothing is the dominant revenue category across all seasons and metrics.
- Outerwear consistently underperforms on revenue, volume, seasonal demand, and stock depth.
- Spring is the strongest overall trading season — optimal for promotional and inventory investment.
- Medium (M) is the top size in every category without exception.
- Accessories provides the most seasonally stable revenue contribution.
- Male customers represent ~68% of the base, but spend identically to female customers.

9.2 Null Findings (Variables That Do Not Predict Spend)

Variable	Hypothesis Tested	Verdict
Age	Older customers spend more	REJECTED — no age-spend relationship
Gender	Gender predicts basket size	REJECTED — identical distributions
Loyalty (Previous Purchases)	Loyal customers spend more	REJECTED — zero correlation
Discounts	Discounts lift basket size	REJECTED — marginal negative effect
Promo Codes	Promos increase spend	REJECTED — no category-level lift
Subscription	Subscribers spend more	REJECTED — identical to non-subscribers
Payment Method	Method correlates with spend	REJECTED — uniform across methods

10. Recommendations

10.1 Immediate Actions

- Investigate Outerwear — audit product range, pricing, and visibility; every chart signals a category in distress.
- Review size S stock levels in Clothing — potential lost sales from L-heavy stocking.
- Re-evaluate promotional ROI — shift discount and promo code measurement from basket size to frequency, retention, and acquisition impact.
- Prioritise spring season inventory and marketing investment — strongest overall trading period.

10.2 Analytical Next Steps

- Conduct CLV (Customer Lifetime Value) analysis to assess whether subscription and loyalty programmes drive long-term frequency gains.
- Build a multivariate model (income, location-weighted by population, category affinity) to identify real spend drivers.
- Validate all findings against real, unbalanced transactional data — current dataset is synthetically generated.
- Segment customers by purchase frequency, not spend, to design targeted retention programmes.
- Analyse promo code cohorts over time to evaluate acquisition and repeat-purchase effects separately.

10.3 Data Quality Recommendations

- Flag this dataset as synthetic for all internal reporting — it should not be the basis for capital allocation decisions.
- Normalise any geographic analysis by state population or retail market size before drawing location-based conclusions.
- Investigate the \$20–\$100 hard pricing boundary — determine whether it reflects real catalogue tiers or a data generation constraint.

Appendix: Analysis Coverage

The following analytical outputs were produced and informed this report:

Analysis	Chart Type	Key Variable(s)
Purchase Amount Distribution	Histogram + KDE	Purchase Amount
Revenue by Category	Bar chart (sum)	Category, Purchase Amount
Gender vs. Spending	Box plot	Gender, Purchase Amount
Age vs. Spending Trend	Scatter + regression	Age, Purchase Amount
Discount Impact on Spend	Bar chart (mean)	Discount Applied, Purchase Amount
Promo Code Impact	Box plot + bar chart	Promo Code Used, Purchase Amount, Category
Payment Method Share	Pie chart + count plot	Payment Method
Seasonal Revenue Heatmap	Heatmap (pivot table)	Category, Season, Purchase Amount
Size Distribution by Category	Count plot	Category, Size

Analysis	Chart Type	Key Variable(s)
Repeat Purchases vs. Spend	Scatter plot	Previous Purchases, Purchase Amount
Subscription vs. Spend	Box plot	Subscription Status, Purchase Amount
Age Distribution	Histogram + KDE	Age
Geographic Revenue (Top 10)	Bar chart	Location, Purchase Amount
Payment Method Count	Count plot	Payment Method
Correlation Matrix	Heatmap	All numeric variables
Category Spending Distribution	Box plot	Category, Purchase Amount

— End of Report —